

How to use these token cutouts

These pages contain the text *The Cat in the Hat* by Dr Seuss. The **red text** above each token shows its prefix—the 1 word before it in the original text.

Training: build the cutouts model

1. **cut out the tokens** from the sheets above along the dotted lines (carefully—keep each prefix-token pair together)
2. **spread them out** on a table, face up, no overlap if possible
3. that's your trained language model—every prefix-token pair from your training text is now a physical cutout on the table

Optional extension: once you've got the hang of it, group the cutouts into piles by their **red prefix label**. Each pile is a “bucket” in computer-science speak—the same trick a hash table uses to find values quickly.

Generation: sample new text

1. **choose a starting prefix**—any word that appears as a **red label** on at least one cutout
2. **write it down**, then make it your “current word”
3. **find candidates**—look across the spread for cutouts whose red prefix label matches your current word
4. **pick one** visually (your eye will land on cutouts more often when there are more of them—that's weighted sampling for free)
5. **write down** the picked cutout's token, then **put it back** in the spread
6. the token you just wrote is your new current word—go back to step 3

Tip: if a prefix has more cutouts in the spread, words following it are more common in the source text, and your eye will pick them more often.

Model statistics

- **Total tokens:** 1832 (1832 kept)
- **Entropy:** 2.76 bits/token — how unpredictable each pick is
- **Perplexity:** 6.8 — effective number of choices per pick

the sun did not shine ☐ it

it was too wet to play ☐ so we

we sat in the house all that cold

cold ☐ cold ☐ wet day ☐ I sat

sat there with Sally ☐ we sat

sat there ☐ we two ☐ and I said

said ☐ how I wish we had

had something to do too wet to

to go out and too cold to play

play ball ☐ so we sat in the house

house ☐ we did nothing at all ☐

☐ so all we could do was to sit

sit sit sit and we did not like

like it not one little bit and

and then something went bump

bump how that bump made us jump

jump we looked then we saw him

him step in on the mat we looked

looked and we saw him the cat in

in the hat and he said to us

, Why do you sit there like

like that I know it is wet and

and the sun is not sunny But

But we can have Lots of good fun

fun that is funny I know some

some good games we could play

, said the cat I know some

some new tricks , said the cat in

in the hat a lot of good tricks

tricks I will show them to you

, your mother will not mind at

at all if I do then Sally and I

I did not know what to say

, our mother was out of the

the house For the day But our

our fish said , no no Make that

that cat go away tell that cat in

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in the hat you do not want to

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to play he should not be here

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here he should not be about

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he should not be here When

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When your mother is out now now

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now have no fear have no fear

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fear said the cat my tricks are

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are not bad said the cat in the

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the hat Why we can have

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have Lots of good fun if you

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you wish with a game that I

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I call up up up with a fish put
335 336 337 338 339 340 341 342

put me down said the fish   this
343 344 345 346 347 348 349

this is no fun at all put me down
350 351 352 353 354 355 356 357

down said the fish  I do not wish
358 359 360 361 362 363 364 365

wish to fall have no fear said the
366 367 368 369 370 371 372

the cat  I will not let you fall 
373 374 375 376 377 378 379 380 381

 I will hold you up high as I
382 383 384 385 386 387 388 389

I stand on a ball  with a
390 391 392 393 394 395 396

a book one one hand and a cup
397 398 399 400 401 402 403

cup on my hat But that is not all
404 405 406 407 408 409 410 411

all I can do said the cat   
412 413 414 415 416 417 418 419 420

 look at me look at me now
421 422 423 424 425 426 427

now said the cat with a cup

cup and a cake on the top of my

hat I can hold up two books

books I can hold up the fish and a

a little toy ship and some milk

milk on a dish and look I can

can hop up and down on the ball

ball But that is not all Oh , no

that is not all look at

at me look at me look at me

me now it is fun to have fun But

But you have to know how I

I	can	hold	up	the	cup	and	
can	hold	up	the	cup	and	the	
518	519	520	521	522	523	524	
the	milk	and	the	cake	I	can	
milk	and	the	cake	I	can	hold	
525	526	527	528	529	530	531	
hold	up	these	books	and	the		
up	these	books	and	the	fish		
532	533	534	535	536	537		
fish	on	a	rake	I	can	hold	the
on	a	rake	I	can	hold	the	toy
538	539	540	541	542	543	544	545
toy	ship	and	a	little	toy	man	
ship	and	a	little	toy	man	and	
546	547	548	549	550	551	552	
and	look	with	my	tail	I	can	
look	with	my	tail	I	can	hold	
553	554	555	556	557	558	559	
hold	a	red	fan	I	can	fan	with
a	red	fan	I	can	fan	with	the
560	561	562	563	564	565	566	567
the	fan	as	I	hop	on	the	ball
fan	as	I	hop	on	the	ball	But
568	569	570	571	572	573	574	575
But	that	is	not	all	Oh	,	no
that	is	not	all	Oh	,	no	
576	577	578	579	580	581	582	583
	that	is	not	all			that
585	586	587	588	589	590	591	592
that	is	not	all				that
585	586	587	588	589	590	591	592
is	what	the	cat	said			
what	the	cat	said				then
594	595	596	597	598	599	600	601
then	he	fell	on	his	head	he	
he	fell	on	his	head	he	came	
602	603	604	605	606	607	608	

came down with a bump From up

there on the ball and Sally

Sally and I we saw all the

the things fall and our fish came

came down too he fell into a

a pot he said do I like this

this Oh no I do not this is

is not a good game said our

our fish as he lit no I do

do not like it not one little bit

bit now look what you did said

said the fish to the cat now

now look 697	look at 698	at this 699	this house 700	house look 701	look at 702	at this 703
this look 704	look at 705	at that 706	that you 707	you sank 708	sank our 709	our toy 710
toy ship 711	ship sank 712	， sank 713	sank it 714	it deep 715	deep in 716	in the 717
the cake 718	cake you 719	。 you 720	you shook 721	shook up 722	up our 723	
our house 724	house and 725	and you 726	you bent 727	bent our 728	our new 729	
new rake 730	rake you 731	。 you 732	you should 733	should not 734	not be 735	
be here 736	here When 737	When our 738	our mother 739	mother is 740	is not 741	
not 。 you 742	。 you 743	you get 744	get out 745	out of 746	of this 747	this house 748
house said 749	said the 750	the fish 751	fish in 752	in the 753	the pot 754	pot 。 But 755
But I 757	I like 758	like to 759	to be 760	be here 761	here 。 Oh 762	Oh 。 I 763
I like 766	like it 767	it a 768	a lot 769	lot said 770	said the 771	the cat 772
in the 774	the hat 775	hat to 776	to the 777	the fish 778	fish in 779	in the 780
the 774	hat 775	to 776	the 777	fish 778	in 779	the 780
						pot 781

pot 782	■ 783	I 784	will 785	not 786	go 787	away 788	■ 789	I 790
do 791	not 792	wish 793	to 794	go 795	and 796	so 797	■ 798	said 799
said 800	the 801	cat 802	in 803	the 804	hat 805	■ 806	so 807	so 808
so 809	■ 810	■ 811	I 812	will 813	show 814			
you 815	another 816	good 817	game 818	that 819	I 820			
I 821	know 822	and 823	then 824	he 825	ran 826	out 827	■ 828	
■ 829	and 830	■ 831	then 832	■ 833	fast 834	as 835	a 836	fox 837
■ 838	the 839	cat 840	in 841	the 842	hat 843	came 844	back 845	
back 846	in 847	with 848	a 849	box 850	■ 851	a 852	big 853	red 854
red 855	wood 856	box 857	■ 858	it 859	was 860	shut 861	with 862	
with 863	a 864	hook 865	■ 866	now 867	look 868	at 869	this 870	
this 871	trick 872	■ 873	said 874	the 875	cat 876	■ 877	take 878	a 879

a	look	then	he	got	up	on	
look	then	he	got	up	on	top	
873		874	875	876	877	878	879
top	with	a	tip	of	his	hat	I
with	a	tip	of	his	hat	I	call
880	881	882	883	884	885	886	887
call	this	game	fun	in	a	box	,
this	game	fun	in	a	box	,	said
889		890	891	892	893	894	895
said	the	cat	in	this	box	are	
the	cat	in	this	box	are	two	
897	898	899	900	901	902	903	904
two	things	I	will	show	to		
things	I	will	show	to	you		
905	906	907	908	909	910		
you	now	,	you	will	like		
now	,	you	will	like	these		
911	912	913	914	915	916		
these	two	things	,	said	the		
two	things	,	said	the	cat		
917	918	919	920	921	922		
cat	with	a	bow	,	I	will	pick
with	a	bow	,	I	will	pick	up
923	924	925	926	927	928	929	930
up	the	hook	,	you	will		
the	hook	,	you	will	see		
931	932	933	934	935	936		
see	something	new	,	two			
something	new	,	two	things			
937	938	939	940	941			
things	,	and	I	call	them	Thing	one
,	and	I	call	them	Thing	one	
942	943	944	945	946	947	948	949
one	and	Thing	two	,	these		
and	Thing	two	,	these	things		
949	950	951	952	953	954		

things will not bite you they want

want to have fun then out of

of the box came Thing two and

and Thing one and they ran to us

us fast they said how do

do you do would you like to

to shake hands with Thing one

one and Thing two and Sally and

and I did not know what to do

so we had to shake hands

hands with Thing one and Thing two

two we shook their two hands

hands □	But	our	fish	said	,	no	no
1033	1034	1035	1036	1037	1038	1039	1040
no	those	things	should	not	be		
those	things	should	not	be	in		
1041	1042	1043	1044	1045	1046		
in	this	house	Make	them	go	they	
1047	1048	1049	1050	1051	1052		
they	should	not	be	here	When	your	
1053	1054	1055	1056	1057	1058		
your	mother	is	not	put	them	out	
1059	1060	1061	1062	1063	1064		
out	put	them	out	said	the	fish	
put	them	out	said	the	fish	in	
1065	1066	1067	1068	1069	1070	1071	
in	the	pot	□	have	no	fear	□
1072	1073	1074	1075	1076	1077	1078	
,	little	fish	,	said	the	cat	
little	fish	□	said	the	cat	in	
1079	1080	1081	1082	1083	1084	1085	
in	the	hat	□	these	things	are	
the	hat	□	these	things	are		
1086	1087	1088	1089	1090	1091		
are	good	things	□	and	he	gave	
good	things	□	and	he	gave		
1092	1093	1094	1095	1096	1097		
gave	them	a	pat	□	they	are	
them	a	pat	□	they	are	tame	
1098	1099	1100	1101	1102	1103	1104	
tame	□	Oh	,	so	tame	they	have
□	Oh	□	so	tame	they	have	
1105	1106	1107	1108	1109	1110	1111	

have come here to play they will

will give you some fun on this

this wet wet wet day now

now here is a game that they

they like said the cat they like

like to fly kites said the cat in

in the hat no not in the house

house said the fish in the pot

they should not fly kites in a

a house they should not Oh

the things they will bump Oh

Oh the things they will hit Oh

Oh ◻ 1198	, ◻ 1199	I ◻ 1200	do ◻ 1201	not ◻ 1202	like ◻ 1203	it ◻ 1204	not ◻ 1205
I ◻ 1206	do ◻ 1207	not ◻ 1208	like ◻ 1209	it ◻ 1210	not ◻ 1211	one ◻ 1212	
little ◻ 1206	bit ◻ 1207	then ◻ 1208	Sally ◻ 1209	and ◻ 1210	I ◻ 1211	saw ◻ 1212	
saw ◻ 1213	them ◻ 1214	run ◻ 1215	down ◻ 1216	the ◻ 1217	hall ◻ 1218	◻ 1219	
we ◻ 1220	saw ◻ 1221	those ◻ 1222	two ◻ 1223	things ◻ 1224			
bump ◻ 1225	their ◻ 1226	kites ◻ 1227	on ◻ 1228	the ◻ 1229	wall ◻ 1230		
their ◻ 1225	kites ◻ 1226	on ◻ 1227	the ◻ 1228	wall ◻ 1229	bump ◻ 1230		
bump ◻ 1231	Thump ◻ 1232	Thump ◻ 1233	bump ◻ 1234	down ◻ 1235	the ◻ 1236		
the ◻ 1237	wall ◻ 1238	in ◻ 1239	the ◻ 1240	hall ◻ 1241	◻ 1242	Thing ◻ 1243	two ◻ 1244
two ◻ 1245	and ◻ 1246	Thing ◻ 1247	one ◻ 1248	they ◻ 1249	ran ◻ 1250	up ◻ 1251	
they ◻ 1252	ran ◻ 1253	down ◻ 1254	on ◻ 1255	the ◻ 1256	string ◻ 1257		
string ◻ 1258	of ◻ 1259	one ◻ 1260	kite ◻ 1261	we ◻ 1262	saw ◻ 1263	mother's ◻ 1264	
mother's ◻ 1265	new ◻ 1266	gown ◻ 1267	her ◻ 1268	gown ◻ 1269	with ◻ 1270	the ◻ 1271	
new ◻ 1272	gown ◻ 1273	her ◻ 1274	gown ◻ 1275	with ◻ 1276	the ◻ 1277		
the ◻ 1278	dots ◻ 1279	that ◻ 1280	are ◻ 1281	pink ◻ 1282	◻ 1283	white ◻ 1284	

white and 1273	and red 1274	red then 1275	then we 1276	we saw 1277	saw one 1278	one kite 1279
kite bump 1280	kite bump 1281	bump on 1282	on the 1283	the head 1284	head of 1285	of her 1286
her bed 1287	bed then 1288	then those 1289	those things 1290	things ran 1291	ran about 1292	about with 1293
with big 1294	big bumps 1295	bumps jumps 1296	jumps and 1297	and kicks 1298	kicks and 1299	and with 1300
with hops 1301	hops and 1302	and big 1303	big thumps 1304	thumps and 1305	and all 1306	all kinds 1307
kinds of 1308	of bad 1309	bad tricks 1310	tricks and 1311	and I 1312	I said 1313	said I 1314
I do 1315	do not 1316	not like 1317	like the 1318	the way 1319	way that 1320	that they 1321
they play 1322	play if 1323	if mother 1324	mother could 1325	could see 1326	see this 1327	this Oh 1328
Oh what 1329	what would 1330	would she 1331	she say 1332	say then 1333	then our 1334	our fish 1335
fish said 1336	said look 1337	look and 1338	and our 1339	our fish 1340	fish shook 1341	shook look 1342
look look 1343	look and 1344	and our 1345	our fish 1346	fish shook 1347	shook look 1348	look look 1349

shook with fear , your mother is
1349 1350 1351 1352 1353 1354

is on her way home do you
1355 1356 1357 1358 1359 1360

you hear Oh , what will she do
1361 1362 1363 1364 1365 1366 1367

do to us what will she say Oh ,
1368 1369 1370 1371 1372 1373 1374 1375

, she will not like it to find us
1376 1377 1378 1379 1380 1381 1382 1383

us this way so do something fast
1384 1385 1386 1387 1388 1389

fast said the fish , do you hear I
1390 1391 1392 1393 1394 1395 1396 1397

I saw her , your mother your
1398 1399 1400 1401 1402 1403

your mother is near so , as fast
1404 1405 1406 1407 1408 1409 1410

fast as you can , Think of
1411 1412 1413 1414 1415 1416

of something to do you will
1417 1418 1419 1420 1421

will have to get rid of Thing one
1422 1423 1424 1425 1426 1427 1428

one and 1429	and Thing 1430	Thing two 1431	two so 1432	so  1433	 as 1434	as fast 1435	fast as 1436
as I 1437	I could 1438	could  1439	 I 1440	I went 1441	went after 1442	after my 1443	
my net 1444	net  1445	 and 1446	and I 1447	I said 1448	said  1449	with my 1450	with my 1451
my net 1452	net I 1453	I can 1454	can get 1455	get them 1456	them I 1457	I bet 1458	bet  1459
 I 1460	I bet 1461	bet  1462	 with 1463	with my 1464	my net 1465	net  1466	 I 1467
I can 1468	can get 1469	get those 1470	those things 1471	things yet 1472	yet then 1473	then I 1474	
I let 1475	let down 1476	down my 1477	my net 1478	net  1479	 it 1480	it came 1481	
came down 1482	down with 1483	with a 1484	a PLOP 1485	PLOP and 1486	and I 1487		
I had 1488	had them 1489	them at 1490	at last 1491	last those 1492	those two 1493		
two things 1494	things had 1495	had to 1496	to stop 1497	stop  1498	 then 1499	then I 1500	
I said 1501	said to 1502	to the 1503	the cat 1504	cat  1505	 now 1506	now you 1507	you do 1508
do as 1509	as I 1510	I say 1511	say  1512	 you 1513	you pack 1514	pack up 1515	up those 1516

those things and you take them

away Oh dear said the cat

you did not like our game

Oh dear what a

a shame what a shame what a

a shame then he shut up the

the things in the box with the

the hook and the cat went

went away with a sad kind of look

look that is good said the

the fish he has gone away

Yes But your mother will

will come she will find this big

big mess and this mess is so big

big and so deep and so tall we

we can not pick it up there is

is no way at all and then Who

Who was back in the house Why

Why the cat have no fear of

of this mess said the cat in

in the hat I always pick up all

all my playthings and so

I will show you another good

good trick that I know then we

we saw him pick up all the

the things that were down and he

he picked up the cake and the

the rake and the gown and

and the milk and the strings

and the books and the dish

dish and the fan and the cup

cup and the ship and the fish

fish and he put them away

then he said that is that

and then he was gone with a

a tip of his hat then our

our mother came in and she said

said to us two , did you have

have any fun tell me what did

did you do and Sally and I did

did not know what to say

should we tell her the things

things that went on there that day

day should we tell her about it

it now , what should we do

do Well what would you

you do if your mother asked you